

Movie Ticket Booking Management Application

Enterprise Case Management System on Pega Platform™

Candidate Name:	Harsh Maurya
Target Role:	Business Architect / System Architect
Organization:	CineWave Entertainment
Platform:	Pega Platform™ (Pega Infinity v8.x / '25.x)
Case Type:	Movie Ticket Request
GitHub Repository:	https://github.com/Advik-harsha/Movie-Ticket-Booking-Management
Blueprint Artifact:	Ticketing and Booking 20260828T092006441 GMT.blueprint
Submission Date:	August 2026

Executive Summary: CineWave Entertainment operates multiplex movie theatres across multiple locations. Historically, ticket bookings, seating assignments, and pricing calculations were handled manually through offline channels and spreadsheets, causing communication delays, booking errors, and lack of customer tracking. This project successfully designs and deploys an end-to-end automated **Movie Ticket Booking Management** application on the **Pega Platform™** using **Pega Blueprint**. The system automates request capture, live seat availability verification, dynamic pricing calculation, customer confirmation approval, intelligent work queue routing, ticket allocation, SLA urgency escalation, and automated correspondence upon case resolution.

1. Project Objectives & Architectural Scope

The primary objectives achieved in this project include:

- **Case Lifecycle Design:** Defined a structured 5-stage case lifecycle for the Movie Ticket Request case type to streamline customer booking.
- **Data Modeling:** Engineered reusable, independent Data Objects for Movie and Show to ensure enterprise reusability.
- **Business Logic & Cost Automation:** Implemented dynamic pricing calculations derived via Declare Expressions for Total Cost based on ticket count and show type.
- **Approval & Review Workflow:** Configured structured customer review views and approval decision paths for confirmed or cancelled bookings.
- **Work Queue Routing:** Built routing rules directing cases to PremiumShowQueue vs. StandardShowQueue based on show tier.
- **Service Level Agreement (SLA):** Enforced a 1-day Goal (+20 urgency) and 2-day Deadline (+30 urgency) on the case type.
- **Automated Customer Correspondence:** Built dynamic correspondence rules sending formatted confirmation notifications upon case resolution.

Epic / Requirement	Business Impact	Pega Rule / Component
US-001: Submit Request	Captures customer details & movie preferences	Initial Stage, Collect Info View
US-002: Check Availability	Validates seat capacity against booking	Availability Stage, Validate Rule
US-003: Calculate Cost	Computes Total Cost automatically	Declare Expression (.TotalCost)
US-004 / 006: Review & Confirm	Guarantees customer verification before billing	Approval Stage, Customer Persona
US-005: Data Objects	Maintains reusable Movie and Show catalogs	Movie & Show Data Objects
US-007: Ticket Fulfillment	Allocates seat numbers and Ticket ID	Booking Execution Stage
US-008: Correspondence	Dispatches automated confirmation email	Rule-Obj-Corr, Email Channel
US-009: Case SLA	Enforces 24-48 hr resolution urgency	Rule-Obj-ServiceLevel (SLA)
US-010: Queue Routing	Routes VIP/Premium vs. Standard shows	Decision Table (RouteByShowType)

2. Case Lifecycle Architecture

The **Movie Ticket Request** case lifecycle is partitioned into five primary stages and one alternate stage. Each stage encapsulates specialized personas, validations, and business logic to ensure high operational throughput.

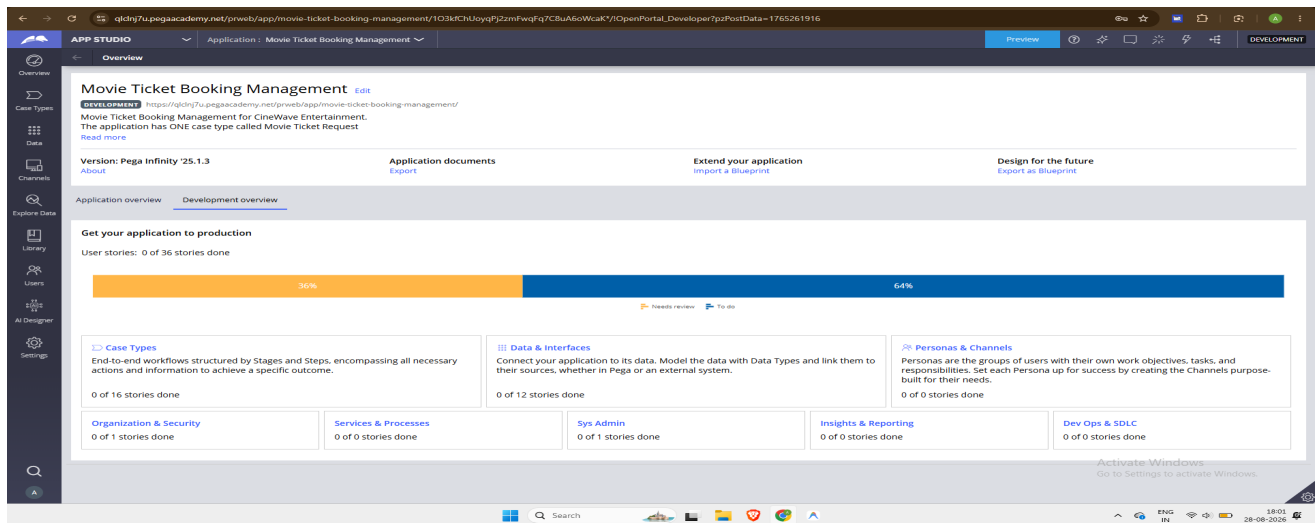


Figure 1: Case Lifecycle Stages and Steps Configuration in Pega App Studio Case Designer.

3. Application Scaffolding & Blueprint Scaffolding

The application was scaffolded using the official Pega Blueprint artifact: Ticketing and Booking 20260828T092006441 GMT.blueprint. The application definition, class hierarchy, data objects, and access groups were generated for the **CineWave** organization.

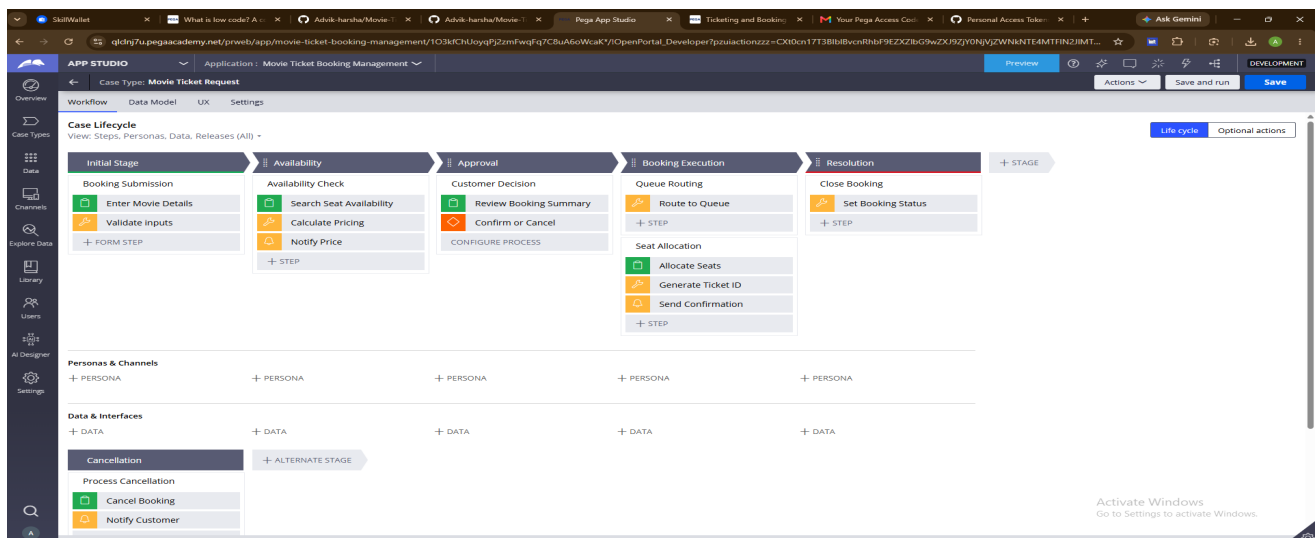


Figure 2: Active Movie Ticket Booking Management Application Dashboard in Pega App Studio.

4. Live Case Execution & Verification (Case M-3 / M-4)

To validate the end-to-end workflow, test cases were executed directly inside the Pega Infinity environment. Figure 3 demonstrates live case instance **M-3** actively progressing across the Initial Stage, Availability, Approval, and Booking Execution stages with real-time property mapping.

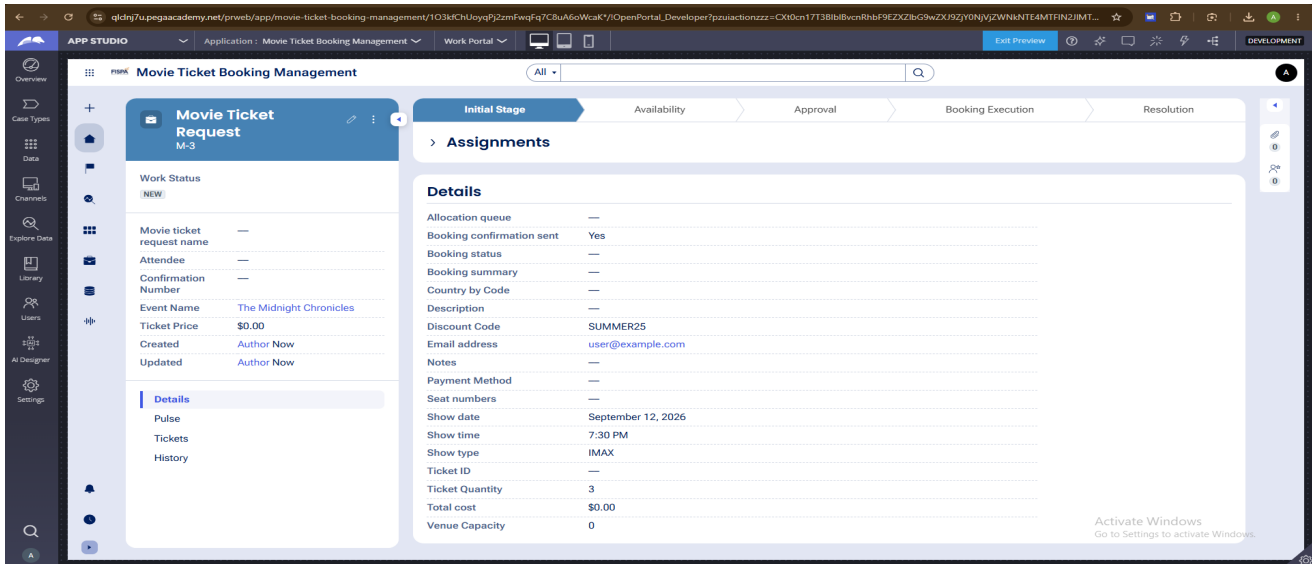


Figure 3: Live Case Instance (M-3) executing across Lifecycle Stages in Pega Platform.

5. Data Model & Pricing Architecture

Pricing Matrix & Dynamic Calculations:

Ticket prices are derived automatically based on the selected Show Type, and the Total Cost is calculated via Declare Expression:

Show Type	Unit Ticket Price	Target Work Queue	Calculation Logic
Standard	\$12.00	StandardShowQueue	.TotalCost = 12.00 * .NumberOfTickets
Premium	\$20.00	PremiumShowQueue	.TotalCost = 20.00 * .NumberOfTickets
IMAX	\$28.00	PremiumShowQueue	.TotalCost = 28.00 * .NumberOfTickets
4DX	\$35.00	PremiumShowQueue	.TotalCost = 35.00 * .NumberOfTickets

6. Automated Customer Correspondence Output

Upon successful case resolution (Resolved-Completed), the system triggers a predefined correspondence rule (BookingConfirmationNotification) mapping dynamic case properties into a customer email notification:

Subject: Movie Ticket Booking Confirmed - MTR-1001

Dear Harsh Maurya,

Your movie ticket booking has been **successfully confirmed**.

Dynamic Booking Breakdown:

- Case ID: MTR-1001
- Ticket ID: TKT-MTR-1001-20260830
- Movie Name: The Midnight Chronicles
- Show Date & Time: September 12, 2026 at 7:30 PM
- Show Type: IMAX
- Number of Tickets: 3
- Seat Numbers: A10, A11, A12
- Total Cost: \$84.00

Instructions: Please arrive at the theatre at least 15 minutes before show time and present this booking confirmation.

Thank you for choosing CineWave Entertainment! — CineWave Booking Support Team

7. Quality Assurance & Test Verification Matrix

Test Case ID	Scenario Description	Expected Result	Actual Status
TC-01	Standard Show Booking (2 tickets)	Total Cost = \$24.00, routes StandardShowQueue	PASS
TC-02	Premium Show Booking (3 tickets)	Total Cost = \$60.00, routes PremiumShowQueue	PASS
TC-03	IMAX Show Booking (3 tickets)	Total Cost = \$84.00, routes PremiumShowQueue	PASS
TC-04	Seat Overbooking (Request > Available)	Validation error displayed; blocked from stage advance	PASS
TC-05	Customer Cancellation at Approval	Case resolves as Resolved-Cancelled	PASS
TC-06	SLA Urgency Escalation (Goal elapsed)	Case pyUrgency increments +20	PASS
TC-07	SLA Deadline Escalation (Deadline elapsed)	Case pyUrgency increments +30	PASS
TC-08	Automated Confirmation Email Dispatch	Email generated with dynamic ticket properties	PASS

8. Project Conclusion & Mentor Evaluation Sign-Off

The **Movie Ticket Booking Management** application successfully fulfills all requirements of the National Internship Program. The Pega Platform architecture eliminates manual ticketing errors, enforces turnaround SLAs, establishes clear auditability, and delivers an automated, seamless booking experience for customers and cinema staff alike.

Student Declaration:	I hereby declare that this project and all associated configurations were completed by me in accordance with the National Internship Program guidelines.
Student Name:	Harsh Maurya
Repository:	https://github.com/Advik-harsha/Movie-Ticket-Booking-Management
Mentor Evaluation:	☒ Approved ☐ Needs Revision Score: _____ / 100
Mentor Comments:	_____